# Mini_projet_1

## ÉTAPE 0 — Préparer le terrain :

```
arij@master:~$ mkdir -p ~/miniproj1/manifests
arij@master:~$ cd ~/miniproj1
arij@master:~/miniproj1$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/arij/miniproj1/.git/
arij@master:~/miniproj1$ git config user.name "arijbelaid"
arij@master:~/miniproj1$ git config user.email "aarijbelaid@gmail.com"
```

## Créer un namespace :

```
arij@master:~/miniproj1$ kubectl create namespace mp1-arij
namespace/mp1-arij created
arij@master:~/miniproj1$ kubectl config set-context --current --namespace=mp1-arij
Context "default" modified.
arij@master:~/miniproj1$ kubectl config view --minify | grep namespace
    namespace: mp1-arij
arij@master:~/miniproj1$ echo "*.bak" > .gitignore
```

## ÉTAPE 1 — Déployer l'API (3 réplicas) :

```
arij@master:~/miniproj1$ kubectl create deployment api \
  --image=ghcr.io/stefanprodan/podinfo:6.5.4 \
  --replicas=3 \
  --dry-run=client -o yaml > manifests/api-deployment.yaml
```

## Modifier manifests/api-deployment.yaml par

```
 GNU nano 7.2                                  manifests/api-deployment.yaml *
apiVersion: apps/v1
kind: Deployment
metadata:
  name: api
  labels:
    app: api
    tier: backend
spec:
  replicas: 3
  selector:
    matchLabels:
      app: api
      tier: backend
  template:
    metadata:
      labels:
        app: api
        tier: backend
    spec:
      containers:
      - name: api
        image: ghcr.io/stefanprodan/podinfo:6.5.4
        ports:
        - containerPort: 9898
```

## Appliquer et vérifier :

```
arij@master:~/miniproj1$ kubectl apply -f manifests/api-deployment.yaml
deployment.apps/api created
```

```
arij@master:~/miniproj1$ kubectl get pods -o wide
NAME                   READY   STATUS    RESTARTS   AGE   IP           NODE      NOMINATED NODE   READINESS GATES
api-67c5f5f47d-bq9nc   1/1     Running   0          56s   10.42.0.39   master    <none>           <none>
api-67c5f5f47d-dgfbj   1/1     Running   0          56s   10.42.1.15   worker1   <none>           <none>
api-67c5f5f47d-tvpnx   1/1     Running   0          56s   10.42.3.16   worker2   <none>           <none>
arij@master:~/miniproj1$ kubectl rollout status deployment/api
deployment "api" successfully rolled out
```

## Commit :

```
arij@master:~/miniproj1$ kubectl rollout status deployment/api
deployment "api" successfully rolled out
arij@master:~/miniproj1$ git add manifests/api-deployment.yaml
arij@master:~/miniproj1$ git commit -m "Ajout du Deployment api (3 replicas, podinfo 6.5.4)"
[master (root-commit) 86483ac] Ajout du Deployment api (3 replicas, podinfo 6.5.4)
 1 file changed, 24 insertions(+)
 create mode 100644 manifests/api-deployment.yaml
```

## ÉTAPE 2 — Déployer le frontend (2 réplicas) :

```
arij@master:~/miniproj1$ kubectl create deployment frontend \
  --image=nginx:alpine \
  --replicas=2 \
  --dry-run=client -o yaml > manifests/frontend-deployment.yaml
```

## Modifier manifests/frontend-deployment.yaml par :

```
  GNU nano 7.2                                manifests/frontend-deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: frontend
  labels:
    app: frontend
    tier: frontend
spec:
  replicas: 2
  selector:
    matchLabels:
      app: frontend
      tier: frontend
  template:
    metadata:
      labels:
        app: frontend
        tier: frontend
    spec:
      containers:
      - name: frontend
        image: nginx:alpine
        ports:
        - containerPort: 80
```

## Appliquer et vérifier :

```
arij@master:~/miniproj1$ kubectl apply -f manifests/frontend-deployment.yaml
deployment.apps/frontend created
arij@master:~/miniproj1$ kubectl get pods -o wide
NAME                        READY   STATUS              RESTARTS   AGE     IP           NODE      NOMINATED NODE   READI
NESS GATES
api-67c5f5f47d-bq9nc        1/1     Running             0          4m45s   10.42.0.39   master    <none>           <none
>
api-67c5f5f47d-dgfbj        1/1     Running             0          4m45s   10.42.1.15   worker1   <none>           <none
>
api-67c5f5f47d-tvpnx        1/1     Running             0          4m45s   10.42.3.16   worker2   <none>           <none
>
frontend-7f4664b7cd-kv82z   0/1     ContainerCreating   0          9s      <none>       worker1   <none>           <none
>
frontend-7f4664b7cd-l8jcq   1/1     Running             0          9s      10.42.3.17   worker2   <none>           <none
>
arij@master:~/miniproj1$ kubectl rollout status deployment/frontend
deployment "frontend" successfully rolled out
```

## Commit :

```
arij@master:~/miniproj1$ git add manifests/frontend-deployment.yaml
arij@master:~/miniproj1$ git commit -m "Ajout du Deployment frontend (2 replicas, nginx:alpine)"
[master 57eec6a] Ajout du Deployment frontend (2 replicas, nginx:alpine)
 1 file changed, 24 insertions(+)
 create mode 100644 manifests/frontend-deployment.yaml
```

## ÉTAPE 3 — Services ClusterIP :

## 3.1 Service API :

```
 GNU nano 7.2                                  manifests/api-svc.yaml *
apiVersion: v1
kind: Service
metadata:
  name: api-svc
spec:
  type: ClusterIP
  selector:
    app: api
    tier: backend
  ports:
  - port: 9898
    targetPort: 9898
```

## 3.2 Service frontend :

```
 GNU nano 7.2                                 manifests/frontend-svc.yaml *
apiVersion: v1
kind: Service
metadata:
  name: frontend-svc
spec:
  type: ClusterIP
  selector:
    app: frontend
    tier: frontend
  ports:
  - port: 80
    targetPort: 80
```

## 3.3 Appliquer et vérifier :

```
arij@master:~/miniproj1$ kubectl apply -f manifests/api-svc.yaml
service/api-svc created
arij@master:~/miniproj1$ kubectl apply -f manifests/frontend-svc.yaml
service/frontend-svc created
```

```
arij@master:~/miniproj1$ kubectl get svc
NAME           TYPE        CLUSTER-IP     EXTERNAL-IP   PORT(S)    AGE
api-svc        ClusterIP   10.43.81.206   <none>        9898/TCP   25s
frontend-svc   ClusterIP   10.43.76.174   <none>        80/TCP     20s
arij@master:~/miniproj1$ kubectl get endpoints
Warning: v1 Endpoints is deprecated in v1.33+; use discovery.k8s.io/v1 EndpointSlice
NAME           ENDPOINTS                                      AGE
api-svc        10.42.0.39:9898,10.42.1.15:9898,10.42.3.16:9898   31s
frontend-svc   10.42.1.16:80,10.42.3.17:80                    26s
```

## 3.4 Test DNS interne :

```
arij@master:~/miniproj1$ kubectl get pods -l app=frontend
NAME                        READY   STATUS    RESTARTS   AGE
frontend-7f4664b7cd-kv82z   1/1     Running   0          8m8s
frontend-7f4664b7cd-l8jcq   1/1     Running   0          8m8s
arij@master:~/miniproj1$ kubectl exec -it frontend-7f4664b7cd-kv82z -- wget -qO- http://api-svc:9898/version
{
  "commit": "33dac1ba40f73555725fbf620bf3b4f6f1a5ad89",
  "version": "6.5.4"
}arij@master:~/miniproj1$kubectl exec -it frontend-7f4664b7cd-kv82z -- nslookup api-svc.mp1-arij.svc.cluster.locall
Server:         10.43.0.10
Address:        10.43.0.10:53


Name:   api-svc.mp1-arij.svc.cluster.local
Address: 10.43.81.206
```

## 3.5.commit :

```
arij@master:~/miniproj1$ git add manifests/api-svc.yaml manifests/frontend-svc.yaml
arij@master:~/miniproj1$ git commit -m "Ajout des Services ClusterIP api-svc et frontend-svc"
[master 2e9833f] Ajout des Services ClusterIP api-svc et frontend-svc
 2 files changed, 24 insertions(+)
 create mode 100644 manifests/api-svc.yaml
 create mode 100644 manifests/frontend-svc.yaml
```

## ÉTAPE 4 — Ingress Traefik :

```
 GNU nano 7.2                                  manifests/ingress.yaml *
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: app-ingress
spec:
  ingressClassName: traefik
  rules:
  - host: web.k3s.local
    http:
      paths:
      - path: /
        pathType: Prefix
        backend:
          service:
            name: frontend-svc
            port:
              number: 80
  - host: api.k3s.local
    http:
      paths:
      - path: /
        pathType: Prefix
        backend:
          service:
            name: api-svc
            port:
```

## appliquer et vérifier :

```
arij@master:~/miniproj1$ kubectl apply -f manifests/ingress.yaml
ingress.networking.k8s.io/app-ingress created
arij@master:~/miniproj1$ kubectl get ingress
NAME          CLASS     HOSTS                       ADDRESS                                          PORTS   AGE
app-ingress   traefik   web.k3s.local,api.k3s.local   192.168.159.10,192.168.159.11,192.168.159.12   80      6s
arij@master:~/miniproj1$ kubectl describe ingress app-ingress
Name:             app-ingress
Labels:           <none>
Namespace:        mp1-arij
Address:          192.168.159.10,192.168.159.11,192.168.159.12
Ingress Class:    traefik
Default backend:  <default>
Rules:
  Host           Path  Backends
  ----           ----  --------
  web.k3s.local
                 /   frontend-svc:80 (10.42.3.17:80,10.42.1.16:80)
  api.k3s.local
                 /   api-svc:9898 (10.42.1.15:9898,10.42.0.39:9898,10.42.3.16:9898)
Annotations:     <none>
Events:          <none>
```

## Configuration côté Windows :

```
127.0.0.1 kubernetes.docker.internal
# End of section
192.168.159.10 web.k3s.local api.k3s.local
```

## Test depuis PowerShell

```
PS C:\WINDOWS\system32> ping web.k3s.local
>>

Envoi d'une requête 'ping' sur web.k3s.local [192.168.159.10] avec 32 octets de données :
Réponse de 192.168.159.10 : octets=32 temps<1ms TTL=64
Réponse de 192.168.159.10 : octets=32 temps<1ms TTL=64
Réponse de 192.168.159.10 : octets=32 temps<1ms TTL=64
Réponse de 192.168.159.10 : octets=32 temps<1ms TTL=64

Statistiques Ping pour 192.168.159.10:
    Paquets : envoyés = 4, reçus = 4, perdus = 0 (perte 0%),
Durée approximative des boucles en millisecondes :
    Minimum = 0ms, Maximum = 0ms, Moyenne = 0ms
```

```
PS C:\WINDOWS\system32> curl.exe http://web.k3s.local
>>
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>

<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br/>
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br/>
For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://f5.com/nginx">f5.com/nginx</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
```

## Commit l'Ingress :

```
arij@master:~/miniproj1$ git add manifests/ingress.yaml
arij@master:~/miniproj1$ git commit -m "Ajout de l'Ingress Traefik pour web et api"
[master bc49c14] Ajout de l'Ingress Traefik pour web et api
 1 file changed, 27 insertions(+)
 create mode 100644 manifests/ingress.yaml
arij@master:~/miniproj1$
```

## ÉTAPE 5 — Mise à jour sans coupure :

## 5.1 Lance la boucle de requêtes :

```
PS C:\Users\arijb> while ($true) { curl.exe -s http://api.k3s.local/version; Write-Host ""; Start-Sleep -Milliseconds 500 }
{
  "commit": "33dac1ba40f73555725fbf620bf3b4f6f1a5ad89",
  "version": "6.5.4"
}
```

## 5.2 Modifier l'image dans le manifeste :

```
        tier: backend
    spec:
      containers:
      - name: api
        image: ghcr.io/stefanprodan/podinfo:6.6.0
        ports:
        - containerPort: 9898
```

## 5.3 Appliquer :

```
arij@master:~/miniproj1$ kubectl apply -f manifests/api-deployment.yaml
deployment.apps/api configured
```

## 5.4. Suivre le rollout :

```
arij@master:~/miniproj1$ kubectl rollout status deployment/api
Waiting for deployment "api" rollout to finish: 1 out of 3 new replicas have been updated...
Waiting for deployment "api" rollout to finish: 1 out of 3 new replicas have been updated...
Waiting for deployment "api" rollout to finish: 1 out of 3 new replicas have been updated...
Waiting for deployment "api" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "api" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "api" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "api" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "api" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "api" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "api" rollout to finish: 1 old replicas are pending termination...
deployment "api" successfully rolled out
```

## 5.5.Vérifier les ReplicaSets :

```
arij@master:~/miniproj1$ kubectl get rs -l app=api
NAME             DESIRED   CURRENT   READY   AGE
api-67c5f5f47d   0         0         0       44m
api-7b989b589    3         3         3       5m7s
```

## 5.6. Vérifier la version final :

```
arij@master:~/miniproj1$ curl -s http://api.k3s.local/version 2>/dev/null || \
  kubectl exec -it $(kubectl get pods -l app=api -o name | head -1) -- wget -qO- http://localhost:9898/version
{
  "commit": "357009a86331a987811fefc11be1350058da33fc",
  "version": "6.6.0"
```

## 5.7. Rollback :

```
}arij@master:~/miniproj1$kubectl rollout undo deployment/api
Warning: resource deployments/api was previously managed with 'kubectl apply'. Rolling back will not update the kubectl.
kubernetes.io/last-applied-configuration annotation, which may cause unexpected behavior on future 'kubectl apply' opera
tions. Consider using 'kubectl apply' with your previous configuration file instead.
deployment.apps/api rolled back
arij@master:~/miniproj1$ kubectl rollout status deployment/api
deployment "api" successfully rolled out
arij@master:~/miniproj1$ kubectl get rs -l app=api
NAME             DESIRED   CURRENT   READY   AGE
api-67c5f5f47d   3         3         3       46m
api-7b989b589    0         0         0       6m37s
```

## 5.8.Vérifier dans PowerShell

```
PS C:\Users\arijb> curl.exe http://api.k3s.local/version
{
  "commit": "33dac1ba40f73555725fbf620bf3b4f6f1a5ad89",
  "version": "6.5.4"
```

## 5.9.commit :

```
arij@master:~/miniproj1$ git add manifests/api-deployment.yaml
arij@master:~/miniproj1$ git commit -m "Mise a jour API vers 6.6.0 puis rollback"
[master e050dc2] Mise a jour API vers 6.6.0 puis rollback
 1 file changed, 1 insertion(+), 1 deletion(-)
arij@master:~/miniproj1$
```

## ÉTAPE 6 — Panne et diagnostic :

## Panne A — Tag d'image inexistant :

## A.1 Introduire la panne :

## Modifier 6.6.0 par 6.5.999

```
        tier: backend
    spec:
      containers:
      - name: api
        image: ghcr.io/stefanprodan/podinfo:6.5.999
        ports:
        - containerPort: 9898
```

```
arij@master:~/miniproj1$ kubectl apply -f manifests/api-deployment.yaml
deployment.apps/api configured
```

## A.2 Observer les symptômes

```
arij@master:~/miniproj1$ kubectl get pods -l app=api
NAME                   READY   STATUS             RESTARTS   AGE
api-67c5f5f47d-4xjxs   1/1     Running            0          24m
api-67c5f5f47d-6tpm9   1/1     Running            0          24m
api-67c5f5f47d-tzssx   1/1     Running            0          24m
api-6b7fd665c-x7sz6    0/1     ImagePullBackOff   0          82s
```

## A.3 Diagnostiquer — ordre des investigations

```
arij@master:~/miniproj1$ kubectl get pods -l app=api
NAME                   READY   STATUS             RESTARTS   AGE
api-67c5f5f47d-4xjxs   1/1     Running            0          24m
api-67c5f5f47d-6tpm9   1/1     Running            0          24m
api-67c5f5f47d-tzssx   1/1     Running            0          24m
api-6b7fd665c-x7sz6    0/1     ImagePullBackOff   0          82s
arij@master:~/miniproj1$ kubectl get pods -l app=api
NAME                   READY   STATUS             RESTARTS   AGE
api-67c5f5f47d-4xjxs   1/1     Running            0          25m
api-67c5f5f47d-6tpm9   1/1     Running            0          25m
api-67c5f5f47d-tzssx   1/1     Running            0          25m
api-6b7fd665c-x7sz6    0/1     ImagePullBackOff   0          2m5s
arij@master:~/miniproj1$ kubectl describe pod <nom-pod-en-erreur>
-bash: syntax error near unexpected token `newline'
arij@master:~/miniproj1$ kubectl get events --sort-by=.metadata.creationTimestamp | tail -20
25m         Normal    Started               pod/api-67c5f5f47d-6tpm9    Container started
25m         Normal    Created               pod/api-67c5f5f47d-6tpm9    Container created
25m         Normal    Scheduled             pod/api-67c5f5f47d-tzssx    Successfully assigned mp1-arij/api-67c5f5f47d-tzssx to maste
r
25m         Normal    Pulled                pod/api-67c5f5f47d-6tpm9    Container image "ghcr.io/stefanprodan/podinfo:6.5.4" already
 present on machine and can be accessed by the pod
25m         Normal    SuccessfulDelete      replicaset/api-7b989b589    Deleted pod: api-7b989b589-5dknz
25m         Normal    Pulled                pod/api-67c5f5f47d-tzssx    Container image "ghcr.io/stefanprodan/podinfo:6.5.4" already
 present on machine and can be accessed by the pod
25m         Normal    Started               pod/api-67c5f5f47d-tzssx    Container started
25m         Normal    SuccessfulDelete      replicaset/api-7b989b589    Deleted pod: api-7b989b589-459f8
25m         Normal    Killing               pod/api-7b989b589-459f8     Stopping container api
25m         Normal    Created               pod/api-67c5f5f47d-tzssx    Container created
6m49s       Normal    TaintManagerEviction  pod/api-67c5f5f47d-tzssx    Cancelling deletion of Pod mp1-arij/api-67c5f5f47d-tzssx
2m26s       Normal    ScalingReplicaSet     deployment/api              Scaled up replica set api-6b7fd665c from 0 to 1
2m24s       Normal    Scheduled             pod/api-6b7fd665c-x7sz6     Successfully assigned mp1-arij/api-6b7fd665c-x7sz6 to worker
2
2m25s       Normal    SuccessfulCreate      replicaset/api-6b7fd665c    Created pod: api-6b7fd665c-x7sz6
```

## A.4 Corriger

## Rendre l'image 6.6.0

```
    containers:
    - name: api
      image: ghcr.io/stefanprodan/podinfo:6.6.0
      ports:
      - containerPort: 9898
```

## Appliquer + vérifier :

```
arij@master:~/miniproj1$ nano manifests/api-deployment.yaml
arij@master:~/miniproj1$ kubectl apply -f manifests/api-deployment.yaml
deployment.apps/api configured
arij@master:~/miniproj1$ kubectl rollout status deployment/api
deployment "api" successfully rolled out
arij@master:~/miniproj1$ kubectl get pods -l app=api
NAME                   READY   STATUS    RESTARTS   AGE
api-7b989b589-8nscd    1/1     Running   0          10s
api-7b989b589-lflcr    1/1     Running   0          13s
api-7b989b589-qg27p    1/1     Running   0          12s
```

## Panne B — Sélecteur de Service erroné

## B.1 Introduire la panne

```
 GNU nano 7.2                                    manifests/api-svc.yaml
apiVersion: v1
kind: Service
metadata:
  name: api-svc
spec:
  type: ClusterIP
  selector:
    app: api-bad
  ports:
  - port: 9898
    targetPort: 9898
```

## B.2.Vérifier le syntaxe :

```
arij@master:~/miniproj1$ kubectl apply -f manifests/api-svc.yaml --dry-run=client
service/api-svc configured (dry run)
arij@master:~/miniproj1$
```

## B.3.Appliquer :

```
arij@master:~/miniproj1$ kubectl apply -f manifests/api-svc.yaml
service/api-svc configured
```

## B.2 Observer les symptômes

```
arij@master:~/miniproj1$ kubectl get endpoints api-svc
Warning: v1 Endpoints is deprecated in v1.33+; use discovery.k8s.io/v1 EndpointSlice
NAME      ENDPOINTS   AGE
api-svc   <none>      74m
arij@master:~/miniproj1$ kubectl describe svc api-svc
Name:                     api-svc
Namespace:                mpl-arij
Labels:                   <none>
Annotations:              <none>
Selector:                 app=api-bad
Type:                     ClusterIP
IP Family Policy:         SingleStack
IP Families:              IPv4
IP:                       10.43.81.206
IPs:                      10.43.81.206
Port:                     <unset>  9898/TCP
TargetPort:               9898/TCP
Endpoints:
Session Affinity:         None
Internal Traffic Policy:  Cluster
Events:                   <none>
arij@master:~/miniproj1$ kubectl get pods --show-labels
NAME                        READY   STATUS    RESTARTS   AGE   LABELS
api-7b989b589-8nscd         1/1     Running   0          11m   app=api,pod-template-hash=7b989b589,tier=backend
api-7b989b589-lflcr         1/1     Running   0          11m   app=api,pod-template-hash=7b989b589,tier=backend
api-7b989b589-qg27p         1/1     Running   0          11m   app=api,pod-template-hash=7b989b589,tier=backend
frontend-7f4664b7cd-kv82z   1/1     Running   0          80m   app=frontend,pod-template-hash=7f4664b7cd,tier=frontend
frontend-7f4664b7cd-l8jcq   1/1     Running   0          80m   app=frontend,pod-template-hash=7f4664b7cd,tier=frontend
```

## B.3.Test depuis PowerShell

```
PS C:\WINDOWS\system32> curl.exe -i http://api.k3s.local/version
HTTP/1.1 503 Service Unavailable
Content-Type: text/plain; charset=utf-8
X-Content-Type-Options: nosniff
Date: Sun, 11 Oct 2026 19:41:51 GMT
Content-Length: 20

no available server
```

## B.4.Test depuis frontend :

```
arij@master:~/miniproj1$ kubectl exec -it frontend-7f4664b7cd-kv82z -- wget -qO- http://api-svc:9898/version
wget: can't connect to remote host (10.43.81.206): Connection refused
command terminated with exit code 1
arij@master:~/miniproj1$
```

## B.5. Corriger la panne

```
  GNU nano 7.2                                          manifests/api-svc.yaml *
apiVersion: v1
kind: Service
metadata:
  name: api-svc
spec:
  type: ClusterIP
  selector:
    app: api
    tier: backend
  ports:
  - port: 9898
    targetPort: 9898
```

## B.6.Appliquer +Tester :

```
arij@master:~/miniproj1$ kubectl apply -f manifests/api-svc.yaml
service/api-svc configured
arij@master:~/miniproj1$ kubectl get endpoints api-svc
Warning: v1 Endpoints is deprecated in v1.33+; use discovery.k8s.io/v1 EndpointSlice
NAME      ENDPOINTS                                            AGE
api-svc   10.42.0.42:9898,10.42.1.19:9898,10.42.3.21:9898   79m
arij@master:~/miniproj1$ kubectl exec -it frontend-7f4664b7cd-kv82z -- wget -qO- http://api-svc:9898/version
{
  "commit": "357009a86331a987811fefc11be1350058da33fc",
  "version": "6.6.0"
}arij@master:~/miniproj1$
```

## Depuis PowerShell

```
PS C:\WINDOWS\system32> curl.exe http://api.k3s.local/version
{
  "commit": "357009a86331a987811fefc11be1350058da33fc",
  "version": "6.6.0"
}
PS C:\WINDOWS\system32>
```

## Commit :

```
arij@master:~/miniproj1$ git commit -m "Ajout du .gitignore + validation du diagnostic des 2 pannes"
[master a56edf3] Ajout du .gitignore + validation du diagnostic des 2 pannes
 1 file changed, 1 insertion(+)
 create mode 100644 .gitignore
```

## Vérification du front sur  navigateur

